

Questa tesi è nata dal desiderio di esplorare un ambito che conoscevo poco.

La curiosità è la mia fiamma perpetua: a volte è un fuocherello, ma resta acceso anche quando i risultati tardano e la strada si complica. Dedico queste pagine a chi custodisce lo stesso fuoco in luoghi dove il futuro è difficile persino da immaginare.

Ringrazio il mio relatore, il professor Rotondi, per il tempo, la pazienza e l'entusiasmo con cui mi ha guidata. Per aver accolto le mie domande e dato fiducia alla mia iniziativa.

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Ad maiora.

Giulia

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1 Introduction

Over the past decade, machine learning has changed what it means to predict. Methods that impose no functional form on the relationship between inputs and outputs, and that improve as data accumulate, have displaced parametric models across domains in which prediction is the objective. Finance has followed, and the question that this movement raises is no longer whether flexible estimators can learn complex relationships (they demonstrably can) but what is gained, and what is given up, when a model derived from economic theory is replaced by one inferred from data. Option pricing is an unusually well-suited setting in which to ask that question. Unlike most prediction problems in finance, it has a strong derived benchmark: the Black–Scholes formula follows from a no-arbitrage argument and a small set of explicit assumptions about the dynamics of the underlying asset. Because the formula rests on assumptions that are stated explicitly, its pricing errors are interpretable: a deviation between formula and market price can be attributed to a particular assumption failing, rather than absorbed into a generic residual. A comparison against it is therefore informative about why a flexible estimator might do better, and not only about whether it does. That is the sense in which this work asks what lies beyond Black-Scholes: not whether the formula should be abandoned, but where exactly it stops being enough. The theoretical case for expecting an improvement is specific. Black–Scholes assumes that the volatility of the underlying is constant across strikes and maturities. Observed markets contradict this directly: prices imply volatilities that vary systematically with both, a regularity that the formula cannot reproduce under any single choice of its inputs. A nonparametric estimator is under no such constraint. If the deviation between market prices and formula prices is systematic rather than random, it is in principle learnable, and the improvement should be largest precisely where the assumption binds hardest. The empirical literature has pursued this for three decades and, read at the level of abstracts and headline figures, appears to have settled the matter in favour of flexible models, read more closely, it has not. Three design choices matter most. The first is whether the competing models are given access to the same information, since several inputs commonly supplied to one side are recovered by inverting a pricing model

against the very price to be predicted. The second is whether the data are partitioned in a way that respects their temporal ordering, since option prices are highly persistent from one day to the next and a random split allows near-duplicates of the same contract to appear on both sides. The third is whether performance is reported in aggregate or conditional on contract characteristics, since an average computed over a population dominated by low-priced contracts can conceal regions of systematic failure. Where these choices are handled loosely, the measured advantage cannot be attributed to the estimator with any confidence. This thesis takes the three requirements as binding constraints, and asks a correspondingly narrower question: under what conditions, and in which regions of the contract space, does a nonparametric estimator price options more accurately than Black–Scholes when both observe exactly the same information? The comparison is run on exchange-traded options on two underlyings with deliberately different characteristics, a single stock and a broad index, over a five-year period. A design of this kind forgoes several specifications that would raise measured accuracy on one side or the other; where the analysis identifies one, it is stated as a direction for further work rather than implemented. Imposing the constraints changes the answer. The advantage of the flexible estimator survives, but it is local rather than general: it is concentrated where theory predicts the parametric assumption to fail, and it reverses elsewhere. The reversal, rather than the advantage, is what this work sets out to explain, that is, why a model free of the parametric constraint does not dominate everywhere. The remainder proceeds as follows. Section 2 sets out the parametric benchmark and its limitations. Section 3 reviews the comparative literature. Section 4 describes the data and the estimation design. Section 5 reports the comparison, Section 6 the structural diagnosis, the findings and their limitations.

2 The Black–Scholes Model and Its Empirical Limitations

2.1 The pricing formula

Black and Scholes (1973) derive the price of a European option from a no-arbitrage argument. Their argument is one of replication: a position in the underlying, financed partly by borrowing, can be made to pay exactly what the option pays in every possible path of the underlying, provided the position is adjusted continuously as the underlying moves. Two portfolios with identical payoffs must have identical prices, or a riskless profit would be available; the price of the option is therefore the cost of building the replicating portfolio. That cost can be computed directly from observable quantities (the price of the underlying and the risk-free rate) with no need to forecast the underlying or to specify investors' attitude to risk. The resulting price is therefore unique. The derivation rests on the following assumptions:

- (A1) the price of the underlying follows a geometric Brownian motion with constant drift and constant volatility;
- (A2) the risk-free rate is known and constant over the life of the option;
- (A3) trading is continuous and the underlying is perfectly divisible;
- (A4) there are no transaction costs, no taxes, and no restrictions on short selling, with full use of the proceeds;
- (A5) the option is European, exercisable only at expiry;
- (A6) the underlying pays no dividends over the life of the option, or pays them at a known continuous yield;
- (A7) no riskless arbitrage opportunities are available.

All of these assumptions are approximations. Section 2.3 examines the four whose failure is both observable in the data used here and consequential for the prices being compared.

Extending the argument to an underlying paying a continuous dividend yield q (Merton, 1973) gives, for a European call,

$$C = Se^{-qT}N(d_1) - Ke^{-rT}N(d_2), \quad (2.1)$$

with

$$d_1 = \frac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}, \quad (2.2)$$

where S is the price of the underlying, K the strike, T the time to maturity, r the risk-free rate, σ the volatility of the underlying, and $N(\cdot)$ the standard normal cumulative distribution function.

Two properties of (2.1) matter for what follows. First, the formula is homogeneous of degree one in (S, K) : scaling both by $\lambda > 0$ leaves $\ln(S/K)$ unchanged, hence leaves $N(d_1)$ and $N(d_2)$ unchanged, and scales the price by λ . Only the ratio S/K , not the two levels separately, carries pricing information. Section 4 returns to this property, which has direct implications for how a nonparametric estimator should be parameterised.

Second, the formula behaves correctly in the limits. As the option moves far into the money, $N(d_1)$ and $N(d_2)$ approach one and the price converges to the discounted intrinsic value $Se^{-qT} - Ke^{-rT}$; far out of the money both terms vanish and the price converges to zero. These limits hold by construction, for any parameter values, and Section 6 shows that this is not a property an estimator fitted to data can be assumed to share.

2.2 Volatility: historical, implied, and the smile

Of the six inputs to (2.1), four are contractual or directly observable. The volatility is not. It is the standard deviation of the underlying's returns over the remaining life of the option, a quantity that is not known when the option is priced and must be estimated.

Two estimators are commonly used. Historical volatility is calculated as the standard deviation of realised log returns over a fixed past window and is then annualised. It is observable ex ante and does not use information from the option market. However, it

is backward-looking: using it as an estimate of future volatility requires the assumption that the recent volatility regime will persist over the remaining life of the option. Implied volatility, by contrast, is obtained by numerically inverting (2.1) using the observed market price of an option. It is forward-looking in the sense that it reflects the market's assessment of future volatility. However, it cannot independently explain the price from which it was obtained: substituting a contract's implied volatility back into the formula reproduces its observed price by construction.

The two estimators also differ in the structure they impose across option contracts. Under a given estimation procedure and on a given date, historical volatility assigns a single volatility value to the underlying asset. The same estimate is therefore used for all options written on that asset, regardless of their strike or maturity. This is consistent with the structure of (2.1), which assumes a single value of σ for the underlying.

Implied volatilities do not exhibit this structure. Inverting the formula for the options quoted on the same date generally produces different implied volatilities across strikes. These values do not vary randomly: when plotted against strike, they form a systematic pattern known as the volatility smile or, when asymmetric, the volatility skew. Implied volatility also varies across maturities, generating a term structure. The market therefore prices options as if the volatility of the same underlying depended on both the strike and the time to maturity of the contract. Consequently, no single value of σ can reproduce the entire cross-section of observed option prices.

2.3 Where the assumptions fail

The failure of (A1) is the most consequential of the four. No single value of σ reproduces a cross-section that requires several: an estimate calibrated to at-the-money contracts misprices those away from the money, and the reverse. The model admits one volatility parameter where the market demands a function of strike and maturity, and the resulting error is therefore systematic and signed.

Three further departures are worth stating, since each bears on the empirical design.

The lognormality embedded in (A1) also understates the frequency of large moves.

On the AAPL daily log returns used in this sample, excess kurtosis is 4.87, well above the value of zero implied by normality. Figure 1 shows this directly: the bulk of the distribution tracks the normal reference line closely, while the extreme quantiles depart from it visibly on both sides, with the largest observed daily move roughly twice the size a normal distribution with the same variance would predict. Large moves occur more often than the model assumes, and this may affect the pricing of far out-of-the-money calls, which require a sufficiently large increase in the underlying price over their remaining life to finish in the money.

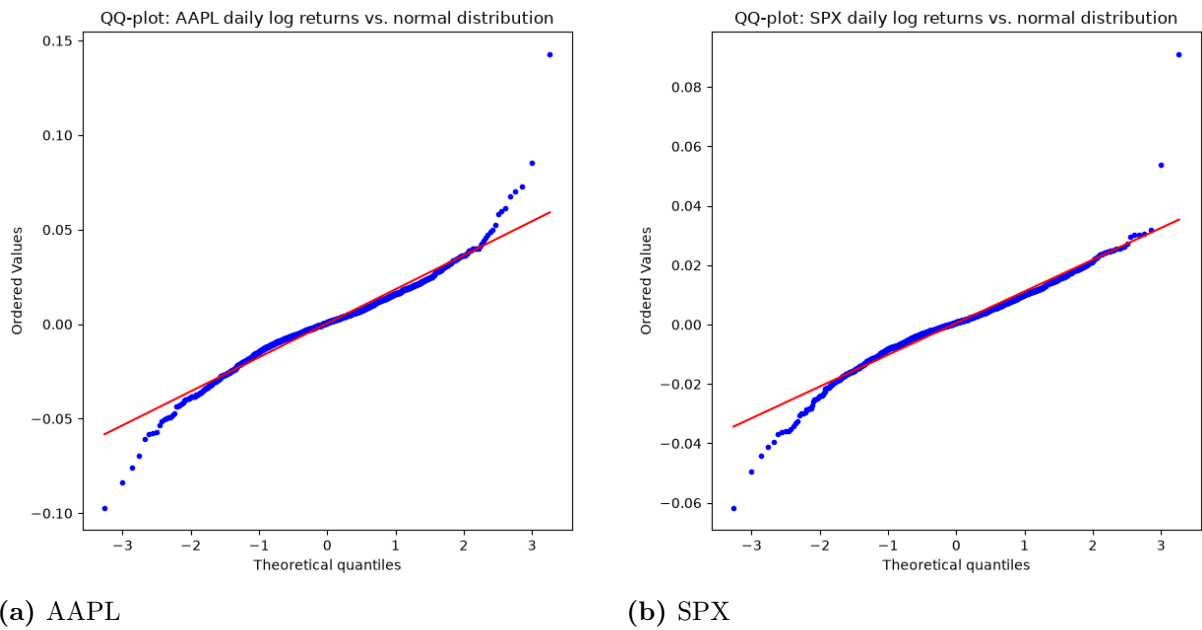


Figure 1: Quantile-Quantile plot of daily log returns against a normal distribution with the same mean and variance. The central quantiles track the reference line closely for both underlyings; the extreme quantiles depart from it on both sides, indicating fatter tails than lognormality allows.

Assumptions (A3) and (A4) do not hold. Rebalancing is discrete rather than continuous, and each adjustment of the replicating portfolio incurs commissions and the cost of crossing the bid-ask spread. Continuous replication is therefore not merely impractical but infinitely costly, and the price in (2.1) is the cost of a portfolio that cannot actually be maintained.

Finally, (A5) holds for one of the two underlyings and not the other. Index options on the S&P 500 are European and the assumption is satisfied exactly. Single-stock options on AAPL are American. The restriction of the sample to calls, adopted in Section 4.2 for unrelated reasons, reduces the importance of this departure. Early exercise of a call on a stock paying no dividends offers no advantage (Hull, 2022, Section 11.5), since exercising

forfeits any remaining time value and requires immediate payment of the strike. On a dividend-paying stock, exercise may become optimal immediately before an ex-dividend date, particularly for in-the-money contracts with little remaining time value. AAPL distributed dividends quarterly at an annual yield below one per cent over the sample period, so the incentive to exercise early is expected to be limited and concentrated around these dates. The early-exercise premium omitted by (2.1) is therefore expected to be small for much of the sample, although its magnitude is not directly estimated here and the right to exercise can affect prices before an ex-dividend date. Omitting this premium lowers the theoretical price relative to an otherwise equivalent American valuation. The contracts most exposed to this effect tend to be deeply in-the-money, the region in which Section 5 finds the formula to outperform the nonparametric models. This approximation should therefore be kept in mind when interpreting the benchmark's advantage in that region, although its effect on the measured pricing error cannot be established without a direct comparison.

3 Literature

3.1 The nonparametric approach to option pricing

Hutchinson et al. (1994) is the standard reference for the alternative approach considered here. Their argument is a direct response to the structure set out in Section 2: because the parametric price follows from an assumed process for the underlying, a misspecification of that process propagates into every price the formula produces. They first train learning networks on data simulated from (2.1) itself, and show that the network recovers the formula without it being imposed, a validation that the method can learn a known relationship before it is asked to learn an unknown one. They then apply the same networks to traded S&P 500 futures options, where the relationship between inputs and price is not assumed to be given by any parametric model.

Two elements of their design are adopted here. The first is the normalisation of the pricing function by the strike, $C/K = f(S/K, \sigma, r, T)$, which implements the degree-one homogeneity discussed in Section 2.1 and reduces the number of inputs the estimator must learn. The second is the restriction of the sample to call options, which keeps contract type unambiguous for both the parametric benchmark and the estimators without introducing an additional categorical feature.

Garcia and Gençay (2000) pursue the same objective by a different route. Degree-one homogeneity is a property the pricing function is known to possess, and an estimator given the underlying price and the strike as separate inputs has to infer it from the data. Rather than normalising the inputs and the target, they build the property into the network itself, separating the pricing function into a component driven by moneyness and one driven by maturity, so that the resulting estimator satisfies homogeneity by construction. They call this supplying the network with a homogeneity hint: a structural regularity taken from theory and imposed on the estimator instead of being left for it to discover. Their comparison is between an estimator that receives the hint and an otherwise identical one that does not, and the constrained estimator generalises better out of sample. The direction of that result is worth noting, because the constrained network is the less flexible

of the two: the gain comes from removing the burden of learning something the theory already supplies, not from added capacity. This architectural route is not implemented here, where homogeneity is imposed only through normalisation of the target and the inputs (Section 4.5); building it directly into the network’s structure, as Garcia and Gençay do, is left as a direction for further work (Section 6.4).

3.2 Comparative evidence and its heterogeneity

The literature comparing nonparametric estimators to (2.1) is large. Ruf and Wang (2020) survey it and count well over a hundred published studies, reporting that more than half of the abstracts they examine emphasise favourable results for neural networks. Read at that level, the question appears settled. The reported magnitudes, however, vary by orders of magnitude, and the variation tracks research design rather than model choice. Three recurring features account for most of it.

The first is the treatment of information. Malliaris and Salchenberger (1993), in one of the earliest comparisons on traded data, supply their network with the previous day’s market price of the option itself alongside the standard inputs (`LAG MARKET PRICE`), while the parametric benchmark receives no such variable. Option prices are highly persistent from one day to the next, so a model given the lagged price can achieve a low error by predicting little change: the exercise becomes one of forecasting a daily increment rather than of valuing a contract. Whenever one side observes a quantity derived from the price to be predicted and the other does not, the measured gap reflects the difference in information as much as the difference in models.

The second is how the sample is split into training and test data. Ivaşcu (2021) compares several nonparametric estimators against (2.1) on European calls written on crude oil futures. He divides the sample at random, deliberately, so that the test set is not concentrated in any single period. However, each contract in that sample is quoted on many consecutive days, at prices that change little from one day to the next. A random split therefore leaves near-copies of the same option on both sides of the division: part of what the estimator meets in testing, it has already seen in training. This is the leakage

that Ruf and Wang (2021) identify. The same paper also runs the comparison a second time, splitting the data chronologically instead of at random: the models are trained on one year of data and tested on the month that follows, with the window rolled forward a month at a time across several consecutive periods. In this second version, the advantage of the nonparametric estimators is much smaller. Nothing about the models changes between the two exercises, only the way the sample is divided.

The third is the region of the contract space over which performance is measured. Studies that report a single aggregate figure cannot reveal whether the ordering between approaches is uniform, and those that disaggregate generally find that it is not. Bennell and Sutcliffe (2004) report the network as clearly superior for OTM contracts, comparable near the money, and weaker for deep ITM and long-maturity ones. Malliaris and Salchenberger (1993) report their results separately for ITM and OTM contracts. The network is more accurate out OTM in four of their five test periods, while the parametric formula is more accurate ITM in three of the five. This is despite an information set that favours the network, since it observes the option's own price from the previous day and the formula does not. Ivaşcu (2021) does disaggregate, by moneyness and by maturity, and reports the nonparametric estimators ahead in every bucket including the deepest ITM one, but the sample is filtered to contracts with moneyness below 1.7 (deep in-the-money) and a premium above one currency unit (deep out-the-money). Conditioning results on moneyness is therefore a necessary step, but on its own it does not guarantee an informative comparison: the range retained by the filters determines which regions can be observed at all.

A fourth element, cutting across the three, is the specification of volatility supplied to the parametric benchmark. Ivaşcu (2021) reports (2.1) under both an implied and a historical estimate on the same sample, obtaining errors of 1.40 and 1.99 respectively: the choice of input accounts for a gap of the same order as some of the differences between competing models. Section 4 states the choice made here and its consequences explicitly.

3.3 Three methodological requirements

Ruf and Wang (2020) close their survey with three recommendations, which this thesis adopts as constraints on the design rather than as refinements to it: inputs should be stationary, the benchmark should be appropriately specified, and the time-series structure of the data should not be violated when partitioning it.

The third is the most consequential and the best documented. Ruf and Wang (2021) show on option data that random partitioning leaks information into training and inflates measured performance, and that a chronological split mitigates it, the same mechanism illustrated in Section 3.2, now confirmed by two papers (Ivaşcu's and Ruf and Wang's).

The first recommendation (stationary inputs) follows from the same homogeneity argument as Hutchinson et al. (1994). Over a multi-year sample the underlying price drifts, and the strikes drift with it, since exchanges open new strikes around wherever the underlying happens to be trading. Their ratio does not drift: traded contracts occupy much the same range of moneyness at the start of the sample as at the end. Moneyness is therefore a stationary input, while the underlying price and the strike, taken separately, are not. An estimator that takes the price and the strike as separate inputs is therefore asked, in the test period, to work with values it never saw in training; one that takes moneyness is not. Section 4 states where this thesis satisfies the requirement and where it does not.

The second requirement is the one this thesis satisfies least completely. Much of the literature evaluates Black-Scholes with implied volatility, and on the evidence reviewed above that specification performs better than the same formula with a historical estimate. It is not adopted here, for the reason given in Section 2.2: a contract's implied volatility is recovered from its own observed price, so supplying it to the formula asked to reproduce that price makes the comparison degenerate. Whether a lagged implied volatility could serve instead, being public information at the time of pricing, is a question this thesis does not settle. The design adopted here gives the parametric formula and the nonparametric estimators the same inputs, at the cost of denying both of them information that would make them more accurate.

4 Data and Methodology

4.1 Sources and sample

Option data are drawn from the OptionMetrics IvyDB US database accessed through Wharton Research Data Services (WRDS) (OptionMetrics, 2026), which records end-of-day quotes for every exchange-traded option on US underlyings¹. Two underlyings are used: AAPL, a single stock, and SPX, a broad equity index. The two are chosen for the differences between them rather than for any similarity: SPX options are European and cash settled; AAPL options are American, physically settled, and written on a stock that pays quarterly dividends. The two underlyings appreciated by a similar amount over the sample, 84.6% for SPX and 79.9% for AAPL, but they differ by more than an order of magnitude in level, and contract prices differ accordingly. Where the two produce different results, the difference is informative about which features of the data drive them. The sample runs from 31 August 2020 to 29 August 2025, covering 1,256 trading days. Four further series are taken from the same source and merged by date: the closing price of the underlying, the dividend yield, historical volatility, and the zero-coupon risk-free curve.

4.2 Filters

The raw files contain 1,994,694 quotations for AAPL and 24,470,893 for SPX. Five filters are applied to the raw quotation files, in the order described below:

Call options only. Put quotations are discarded. A nonparametric estimator has to be fitted separately to calls and to puts, since the two have different payoff structures, so retaining both would require either two estimators or an additional categorical input. The restriction also matters for the treatment of American exercise in Section 2.3: the early-exercise premium that (2.1) omits is negligible for calls, but would not be for puts.

Crossed quotes. Observations where the bid exceeds the ask are removed. Such a quote cannot arise in a functioning market.

¹The code used for data preparation, estimation and evaluation is available at <https://github.com/glmarcant/Priced-by-intelligence>; the option data are proprietary and are not included.

Non-positive bid. A zero bid means no buyer is quoting for the contract, so only one side of the market exists. The midpoint of a one-sided quote is half the ask, which measures nothing about the contract's value.

Untraded contracts. Observations with zero volume and zero open interest on the quotation date are removed. A contract that neither traded nor is held by anyone has a quote that no participant has acted on.

Same-day expiry. Contracts expiring on the quotation date are removed, since $T=0$ leaves $d1$ and $d2$ in (2.2) undefined.

After the final filter, a check confirms that no remaining observation has a non-positive strike, underlying price, volatility or option price, and that no model input is missing after the merges described below; no observation is removed at this stage. The estimation samples contain 838,276 observations on AAPL and 7,688,150 on SPX.

No minimum-price filter is applied, no minimum maturity is imposed beyond $T>0$, and no restriction is placed on moneyness. All three are common in this literature, and each would remove part of the region in which the parametric and nonparametric approaches are most likely to differ.

4.3 Target and inputs

The target is the mid-quote, $(best_bid + best_offer)/2$. No transaction takes place at this value: it is a convention adopted because quotes are two-sided and a single observed price does not exist. Part of the error attributed to any model is therefore measurement error in the target rather than model error, and this component is common to all models compared.

Every model receives the same five or six inputs: moneyness S/K or S and K separately, time to maturity T in years, the risk-free rate r , the dividend yield q , and historical volatility σ . Moneyness enters as a ratio rather than as two separate levels, for the homogeneity reason set out in Section 3.1; the strike price K is rescaled beforehand, since OptionMetrics stores it multiplied by 1000. Maturity is computed as calendar days between the quotation date and expiry, divided by 365. These five inputs are what

remains, of everything downloaded from OptionMetrics, once fields that are constant, purely identifying, or used only to filter the sample are set aside; Appendix A lists every field in the raw extracts and the reason each was kept or dropped.

Two quantities are deliberately excluded. Implied volatility is not supplied: a contract's implied volatility is obtained by inverting (2.1) against that contract's own observed price, so supplying it to a model asked to reproduce that price makes the exercise degenerate. Lagged option prices, and any transformation of them, are excluded on the same grounds (Section 3.2). The resulting information set is identical across models, which is what allows a difference in measured accuracy to be attributed to the estimator rather than to what each estimator was allowed to see. Historical volatility, as defined in Section 2.2, is taken from the IvyDB `Historical_Volatility` file, which reports it for trailing windows of 10 to 1825 calendar days. The 60-day window is used throughout, following Hutchinson et al. (1994), and is close to the median maturity in the sample (79 days for AAPL and 46 days for SPX). Maturities in the sample range from 1 to 365 day for both underlyings, so for much of it the window is not aligned with the life of the contract being priced; matching the window to maturity is an alternative documented in the literature.

The risk-free rate is taken from the IvyDB zero-coupon curve (Table 12) at the tenor closest to 90 days. When (A2) is relaxed to allow a deterministic term structure, the appropriate rate for (2.1) is the continuously compounded zero-coupon rate corresponding to the option's maturity, which represents the average instantaneous rate over its remaining life. The approximation lies in using a fixed tenor rather than matching the rate to each option's maturity. Because r enters the formula multiplied by T , the pricing effect of this approximation is generally more limited in absolute terms for short-dated contracts, but may become more important at longer maturities. Its magnitude depends on the difference between the selected and maturity-matched rates, as well as on the characteristics of the option.

The dividend yield is constructed differently for the two underlyings, because the source data differ. For SPX, IvyDB reports a daily index dividend yield, which is merged directly by date. For AAPL, the file records discrete declared distributions; regular dividends are retained, amounts declared before the four-for-one split of 31 August 2020 are rescaled

accordingly, and each observation is assigned the most recent declaration on or before its quotation date, annualised by the declared payment frequency. The resulting yield averages 0.42% over the sample, which is the figure underlying the treatment of early exercise in Section 2.3.

4.4 Partitioning and model selection

The sample is partitioned chronologically. [Train: 2020-08-31 to 2024-03-27; Validation: 2024-03-28 to 2024-12-26 Test: 2024-12-27 to 2025-08-29]. No observation from a later date is used to fit a model evaluated on an earlier one, the reason is given in Section 3.3.

Model selection is carried out by rolling-window cross-validation within the training period [24-month training window, six-month validation window, six-month step, five folds]. The configuration minimising validation error is retained and the test set is evaluated once, after selection is complete. The full hyperparameter search grid and the selected configurations are reported in Table 14 to make the model selection procedure transparent.

4.5 Models

Three models are compared.

Black–Scholes. The price is computed directly from (2.1), with the Merton adjustment for a continuous dividend yield, which replaces S with Se^{-qT} . It is not fitted and has no free parameters beyond those supplied.

Gradient-boosted trees. The tree ensemble is implemented with XGBoost. Two choices define the specification: whether the target is the price in dollars or the price divided by the strike, and whether it is transformed by a logarithm.

The four resulting configurations are estimated on both underlyings and compared in Section 6; the normalised, log-transformed one is used as the main model in Section 5.

The two dollar-scale configurations take the strike as a sixth input, in addition to the five listed above. This is necessary rather than optional: moneyness, maturity, the rate,

Table 1: The four specifications

	Inputs	Target	Prediction
Dollar scale	$S/K, K, T, r, q, \sigma$	C	$\hat{y}$
Dollar scale, log	$S/K, K, T, r, q, \sigma$	$\log(1 + C)$	$e^{\hat{y}} - 1$
Normalised	$S/K, T, r, q, \sigma$	C/K	$K\hat{y}$
Normalised, log	$S/K, T, r, q, \sigma$	$\log(C/K)$	$Ke^{\hat{y}}$

$\hat{y}$ is the value the model returns; the last column is the transformation applied to obtain a price in dollars. The same four specifications are estimated for the neural network.

the dividend yield and the volatility are all unchanged when S and K are scaled by the same factor, so a model fitted to C from those five inputs alone would be forced to assign the same price to two contracts with the same moneyness written on very different price levels. The strike restores the information on scale, at the cost of an input that is not stationary over the sample: the underlying rises by +79.9% for AAPL and +84.6% for SPX between the first and the last day and a large share of the test period lies above the range of strikes and prices seen in training, where a tree cannot extrapolate.

The two normalised configurations do not need the strike. The model is fitted to a function of C/K and the prediction is multiplied back by K , so the scale enters after estimation rather than being learned. This imposes homogeneity of degree one in (S, K) by construction, the same property the Black–Scholes price satisfies, and it keeps the target in a comparable range as the underlying rises.

The logarithm changes what the loss penalises. Prices range from a few cents for deep OTM contracts to hundreds of dollars for AAPL and thousands for SPX. On the untransformed scale, a squared-error loss weights a given relative error far more heavily for expensive contracts: a 10% error on a \$100 option contributes 250 times as much as the same relative error on a \$0.20 option. On the log scale the two contribute equally. The logarithm also guarantees positive predictions, since the inverse transformation is an exponential, whereas the untransformed specification produces 5552 negative predicted prices for AAPL, which are not admissible values for a call.

For the dollar-scale target the transformation is $\log(1 + C)$ rather than $\log C$. A pure logarithm makes the loss relative also for contracts quoted at a few cents, where the minimum tick of one cent is a large fraction of the quote. An option whose mid-quote

moves from \$0.01 to \$0.03 without any change in its fundamentals is, on the $\log C$ scale, as far from its starting point as an option moving from \$100 to \$300; on the $\log(1 + C)$ scale it is as far as one moving from \$100 to \$102. For the normalised target the constant is not added.

Hyperparameters are selected with the rolling-window cross-validation of Section 4.4 on the normalised log specification, using the root mean squared error on the price scale as criterion. Because the error varies substantially across folds, the selection rule is not the minimum mean error alone: among the configurations whose mean RMSE is within 1% of the lowest, the one with the smallest standard deviation across folds is retained. The grid covers the maximum tree depth, the learning rate, the number of trees and the subsampling rate, in 72 combinations for AAPL and 18 for SPX; the selected configurations are reported in Table 14. The same hyperparameters are used for all four configurations, so that the comparison isolates the target and the inputs; the final models are refitted on the whole training and validation period and evaluated once on the test set.

Neural network. The network is a feed-forward model with a single hidden layer and logistic activation, implemented with scikit-learn. It is estimated on the same four configurations as the tree ensemble, so that the effect of the target and of the inputs can be compared across the two model families. Inputs are standardized with means and standard deviations computed on the training data only; for the two untransformed targets the target is standardized as well, since the logistic activation does not converge on values that range over several orders of magnitude. The weights are estimated with the Adam optimiser in mini-batches of 2,000 observations, with an L_2 penalty of 10^{-4} and at most 2,000 epochs; training stops when the loss on a randomly drawn 10% of the training data does not improve for 15 consecutive epochs. All remaining settings, including the initial learning rate, are left at the library defaults.

The only hyperparameter selected is the number of hidden units, which with a single layer governs the capacity of the model. Networks with 1 to 20 units are fitted on the training set and compared on the validation set by root mean squared error on the price scale. Unlike the tree ensemble, the network is not tuned by rolling-window cross-validation:

each candidate requires a full training run, and five folds per candidate is not feasible within the computational budget, so the selection uses the single chronological validation split described in Section 4.4. The selected values are 10 units for AAPL and 10 for SPX. The final network is refitted on the training and validation sets together and evaluated once on the test set.

4.6 Evaluation

Performance is reported as mean absolute error on the price in dollars, both in aggregate and conditional on moneyness. Buckets are defined on S/K with thresholds at 0.8, 0.95, 1.05, 1.2, 2 and 5, giving seven regions from deep out-of-the-money to extreme in-the-money. Aggregate figures are reported for comparability with the literature but are not the basis of any conclusion drawn here: the median contract price varies by three orders of magnitude for AAPL and four for SPX across buckets, so an aggregate error measure is dominated by whichever contracts are numerous and expensive. The same applies in the opposite direction to relative measures: the mean absolute percentage error is not reported, because for contracts quoted at a few cents the denominator is close to zero and the average is driven by a handful of observations.

The comparison is conducted at two levels. The four configurations of each model are compared on the validation set, together with the corresponding training error, which measures how much of the fit survives outside the estimation period. Only the selected configuration of each model is refitted on the training and validation sets together and evaluated on the test set, which is used once. Black–Scholes requires no estimation and is therefore evaluated on the same sets without refitting.

Conclusions are drawn from the magnitude of the errors by bucket, and only where the differences are large relative to the spread across buckets and consistent across the two underlyings. This is the appropriate standard for a panel of this kind: the same contract is quoted on many consecutive days and contracts written on the same underlying share the same shocks, so the effective number of independent observations is far smaller than the number of rows, and small differences between models carry little information.

5 Results

5.1 Pricing accuracy across moneyness

Black–Scholes provides the benchmark against which both models are measured, and its own error is not uniform across the sample. Its mean absolute error on the AAPL test set falls from \$2.09 for deep out-of-the-money contracts to \$0.28 for extreme in-the-money ones, while the median price in those two buckets is \$0.20 and \$193: the benchmark error is ten times the price of the contract in the first case and about one and a half thousandths of it in the second. Black–Scholes is accurate where the price is almost entirely intrinsic value and there is little left to estimate, and least accurate where the price consists of time value alone.

Both final machine-learning models are estimated in the specification selected in Section 4.5: five inputs and a $\log(C/K)$ target, with the hyperparameters reported there. Each is refitted on the training and validation periods together and evaluated once on the test set.

Table 2: Mean absolute error by moneyness bucket — AAPL test set

Moneyiness bucket	n	Median price	Black–Scholes	XGBoost	Neural network
Deep OTM	26,376	0.20	2.09	0.20	0.20
OTM	22,324	1.47	2.64	0.57	0.86
ATM	13,178	10.36	2.83	1.18	3.58
ITM	15,059	29.40	2.22	0.88	8.68
Deep ITM	31,571	71.65	1.11	1.68	17.35
Very deep ITM	11,824	137.05	0.32	4.65	19.47
Extreme ITM	4,885	193.45	0.28	79.83	101.52
All	125,217	24.80	1.80	4.35	11.789

Deep OTM ($S/K \leq 0.8$), OTM ($0.8 < S/K \leq 0.95$), ATM ($0.95 < S/K \leq 1.05$), ITM ($1.05 < S/K \leq 1.2$), deep ITM ($1.2 < S/K \leq 2$), very deep ITM ($2 < S/K \leq 5$) and extreme ITM ($S/K > 5$).

Tables 2 and 3 report the mean absolute error of the three models by moneyness bucket. The gradient-boosted model is the most accurate of the three in the first four buckets on AAPL, while the neural network is more accurate than the benchmark only in the cheapest two buckets, deep out-of-the-money and out-of-the-money. From a moneyness of 1.2 onwards Black–Scholes is the most accurate model in every remaining bucket, and

Table 3: Mean absolute error by moneyness bucket — SPX test set

Moneyness bucket	n	Median price	Black–Scholes	XGBoost	Neural network
Deep OTM	30,051	0.43	38.57	1.44	1.34
OTM	227,962	7.25	55.43	11.50	13.95
ATM	490,931	140.90	46.33	16.83	58.78
ITM	251,452	642.05	41.96	13.35	243.58
Deep ITM	123,621	1518.20	17.31	46.31	313.94
Very deep ITM	17,799	3793.35	3.92	265.97	473.48
Extreme ITM	8,528	5294.73	3.60	1394.95	1876.93
All	1,150,344		42.88	31.86	136.11

Deep OTM ($S/K \leq 0.8$), OTM ($0.8 < S/K \leq 0.95$), ATM ($0.95 < S/K \leq 1.05$), ITM ($1.05 < S/K \leq 1.2$), deep ITM ($1.2 < S/K \leq 2$), very deep ITM ($2 < S/K \leq 5$) and extreme ITM ($S/K > 5$).

the gap widens with moneyness in both underlyings, reaching a factor of 286 on AAPL and 387 on SPX in the extreme in-the-money bucket and even more with neural networks. Which model prices more accurately therefore depends on where in the cross-section the question is asked.

The reversal is the central feature of the comparison and it is systematic rather than an artefact of a few contracts: the benchmark’s advantage then grows with moneyness across all three remaining buckets, which together account for 39% of the test observations for AAPL and 13% for SPX, so the result is a property of a large part of the cross-section rather than of a few extreme contracts. The reason for this failure is examined separately in Section 6.

In aggregate Black–Scholes has the lowest error on AAPL, and the gradient-boosted model has the lowest one on SPX; the neural network has the highest aggregate error on both underlyings. None of these figures says much about the models: the aggregate is an average over all contracts, in which each bucket weighs in proportion to its size, and buckets whose median prices differ by three orders of magnitude, so it reflects the composition of the sample as much as the behaviour of the models.

The degradation from the estimation period to the test set is fairly uniform across buckets for the gradient-boosted model, and markedly less so for the neural network, whose test-to-train ratio varies considerably more across the cross-section (Table 4).

Table 4: Estimation-period and test-set error ratio by moneyness bucket

Moneyness bucket	XGBoost			Neural Network		
	Train MAE	Test MAE	Ratio	Train MAE	Test MAE	Ratio
<i>AAPL</i>						
Deep OTM	0.135	0.204	1.51	0.203	0.200	0.98
OTM	0.358	0.567	1.59	0.491	0.863	1.76
ATM	0.689	1.182	1.72	1.518	3.576	2.36
ITM	0.572	0.878	1.53	3.722	8.677	2.33
Deep ITM	1.017	1.677	1.65	5.939	17.347	2.92
Very deep ITM	3.099	4.648	1.50	9.350	19.474	2.08
Extreme ITM	43.759	79.830	1.82	71.192	101.521	1.43
All	1.435	4.350	3.03	4.393	11.789	2.68
<i>SPX</i>						
Deep OTM	0.936	1.437	1.54	1.104	1.337	1.21
OTM	5.071	11.504	2.27	6.295	13.947	2.22
ATM	10.281	16.853	1.64	18.188	58.783	3.23
ITM	10.899	13.354	1.23	55.040	243.583	4.43
Deep ITM	42.540	46.306	1.09	73.342	313.938	4.28
Very deep ITM	199.364	265.967	1.33	302.639	473.482	1.56
Extreme ITM	1,123.772	1,394.954	1.24	1,398.661	1,876.926	1.34
All	26.063	31.862	1.22	47.850	136.108	2.84

Train refers to the combined training and validation periods; errors are in US dollars for AAPL and in index points for SPX; the ratio is the test error divided by the training error. Sample sizes by bucket are as in Table 2 and Table 3.

6 Discussion

6.1 Why the models fail in the money

Overfitting is present but is not what drives the failure documented in Section 5.1. On extreme in-the-money contracts the gradient-boosted model has a mean absolute error of \$43.76 on the data it was estimated on, against \$79.83 out of sample, and of 1,124 against 1,395 index points on SPX: on both underlyings the degradation from one to the other is in line with every other bucket. The neural network shows the same pattern in a stronger form. Its in-sample error exceeds the benchmark’s error from the in-the-money bucket onward on both underlyings, and in the extreme in-the-money bucket its test-to-train ratio is among the lowest in the cross-section (Table 4). A model that cannot fit a region of the sample it has seen is not failing to generalise; it is failing to represent that region at all.

The reason lies in what the price of a deep in-the-money call consists of. Once the underlying is well above the strike, the option is worth approximately its intrinsic value $S - K$, and it is also verified empirically: in the AAPL test set the correlation between the two is 0.991. Pricing these contracts therefore means reproducing a quantity that grows without bound with the level of the underlying. Black–Scholes does this by construction, since S enters the formula directly. A regression tree cannot: its prediction is always an average of target values observed during training, so it cannot return a value above the highest one it has seen.

This ceiling does not arise because the test set contains input values absent from training. For AAPL, the maximum underlying price, strike, and moneyness in the test set all fall within the range observed in the combined training and validation period. What shifts instead is the level of the distribution: the median underlying trades at \$157.96 in training and validation against \$214.00 in the test set, a rise of 35%, while the median moneyness is nearly unchanged (1.08 against 1.06). In the deep in-the-money bucket alone, the median option price rises from \$67.68 to \$91.00.

Combined with the sparsity of that region in training (fewer than 1% of the sample lies beyond moneyness 5) a regression tree has little incentive to place further splits there,

so the contracts that fall in it collapse into a small number of large terminal leaves with an almost constant prediction. Figure 2 shows the resulting error pattern on the AAPL test set: Black–Scholes residuals are centred near zero for most contracts, with a widening tail of large negative values (overpricing) that does not grow monotonically with intrinsic value, whereas the gradient-boosted model’s residuals are almost entirely positive and grow approximately linearly with intrinsic value, the signature of a constant prediction applied across a range of contracts whose true price keeps rising. Same phenomenon verifies for SPX (Figure 3) .

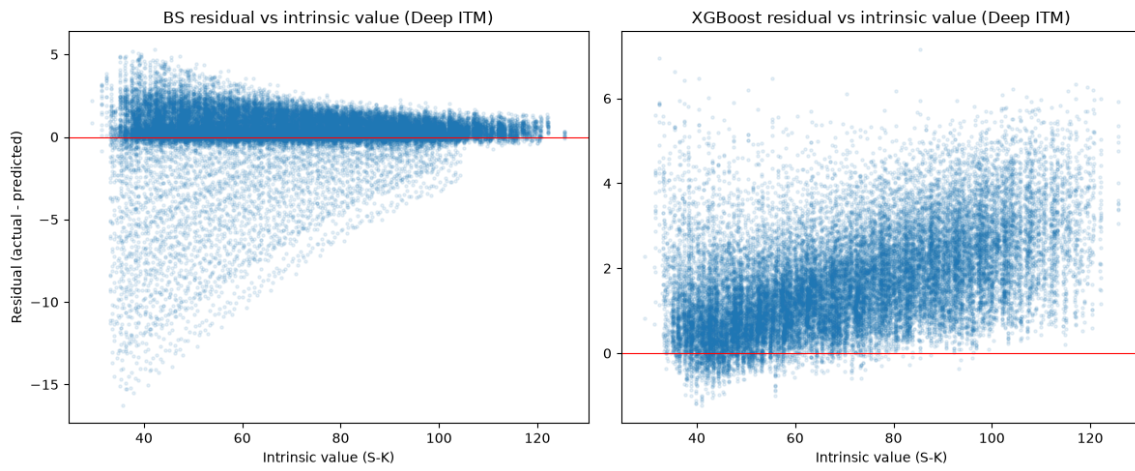


Figure 2: AAPL test set. Residuals (actual – predicted) against intrinsic value ($S - K$) in the deep in-the-money bucket.

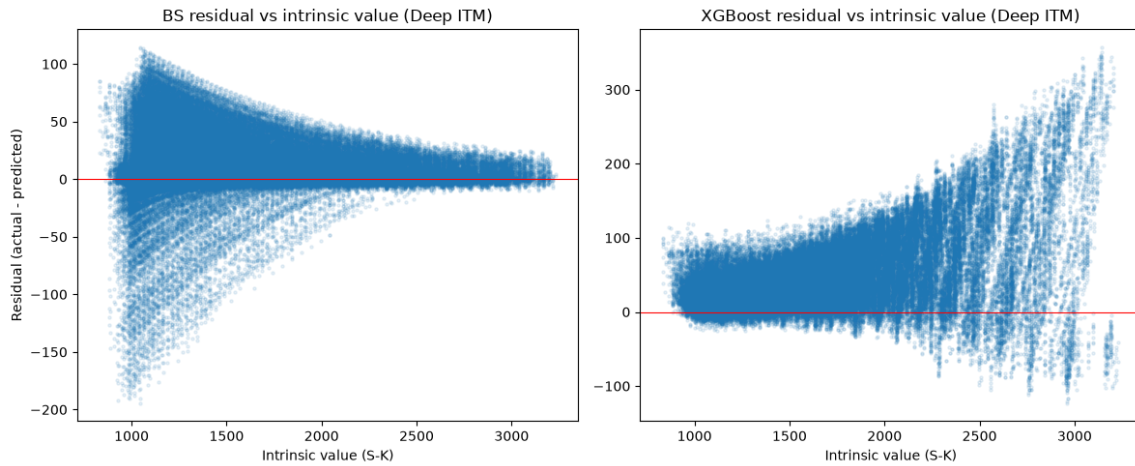
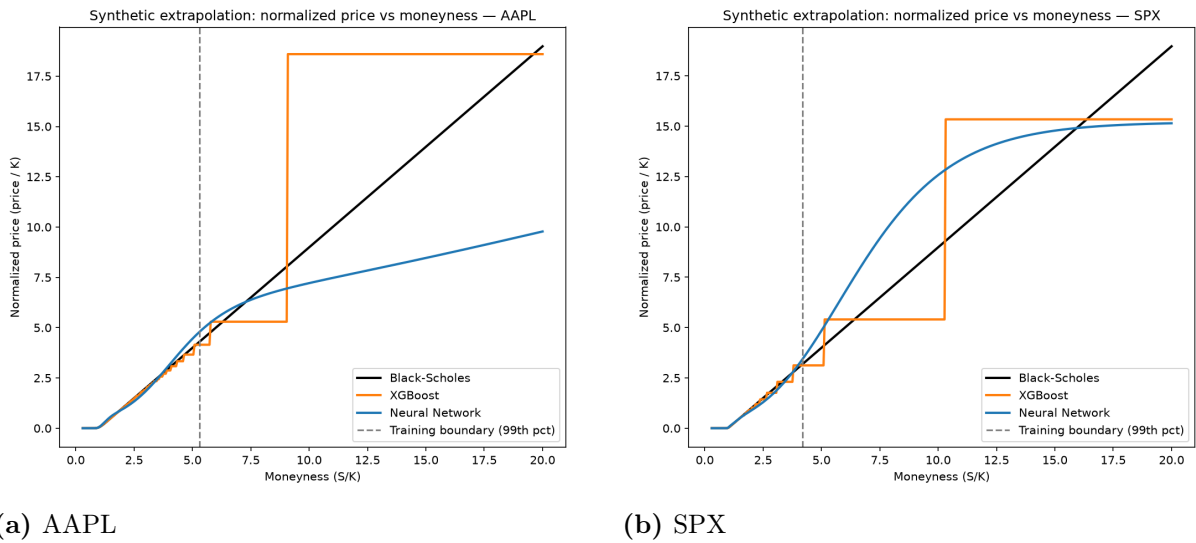


Figure 3: SPX test set. Residuals (actual – predicted) against intrinsic value ($S - K$) in the deep in-the-money bucket.

On SPX the picture combines both mechanisms. The underlying itself partly leaves the training range: 31% of the test observations have a closing price above the maximum

observed in training and validation, against none for AAPL. At the same time, the same level shift documented for AAPL is present here too: the median underlying price rises from \$4,396 to \$5,977, and the median price of a deep in-the-money contract rises from \$1,354 to \$1,668, while the median moneyness is essentially unchanged (1.02 against 1.01). Training-sample sparsity is, if anything, more pronounced than for AAPL: only 0.8% of the training sample has moneyness above 5, and 14.3% above 1.2. The tree ensemble therefore faces both a genuine extrapolation problem on the level of the underlying and the same distributional shift within the region it has already seen.



(a) AAPL

(b) SPX

Figure 4: Synthetic extrapolation beyond the training range. Normalized price (C/K) as a function of moneyness, with the other inputs fixed at their sample medians. The dashed line marks the 99th percentile of training moneyness (5.3 for AAPL, 4.2 for SPX).

Figure 4 makes the resulting mechanism visible. The grid is constructed by fixing the underlying at a representative level for each market (\$200 for AAPL, 4,274.0 index points for SPX) and varying the strike to produce moneyness values from 0.3 to 20 over 400 equally spaced points, so that moneyness increases through the strike, exactly as it does in the data, rather than through the level of the underlying. The remaining four inputs are held fixed at a representative point: time to maturity of 71 days for AAPL and 49 days for SPX, with the risk-free rate, dividend yield and historical volatility set at the corresponding values. Black-Scholes is computed directly from (2.1); XGBoost and the neural network are the final models of the normalised, log-transformed specification (Section 4.5), refitted on the combined training and validation periods. On a synthetic

grid of moneyness, with the remaining inputs held fixed, the Black–Scholes price grows linearly once the option is deep ITM. The tree ensemble tracks it closely up to the training boundary and then departs from it discontinuously: its normalized prediction takes only four distinct values across the whole grid for AAPL and three for SPX, the last of which is returned for every point beyond a fixed moneyness. This is the direct consequence of the tree’s structure: a regression tree partitions the input space into a finite number of regions of constant prediction, and with so few observations beyond that boundary, no further split is warranted. The neural network does not show this discontinuity, its predictions are a continuous function of the inputs, but it does not track Black–Scholes beyond the training boundary either: on AAPL it flattens progressively below the Black–Scholes line, reaching a value of 9.8 against 19.1 at moneyness 20, while on SPX it temporarily overshoots the line between moneyness 5 and 16 before flattening at roughly 16. This pattern is consistent with the saturation of the logistic activation function used in the hidden layer, which bounds each hidden unit’s output between 0 and 1 and would therefore force the network’s prediction toward a constant as moneyness grows, regardless of the linear behaviour the data would imply beyond the observed range; whether this fully accounts for the overshoot on SPX has not been verified directly. The two model families therefore behave differently beyond the training range. Section 6.2 examines how the four specifications compare on this failure.

6.2 What the four specifications show

Section 4.5 introduced four specifications, estimated for both model families: $v1$, with six inputs including the strike and the raw price as target; $v2$, identical in inputs but with $\log(1 + C)$ as target; $v3$, with five inputs (moneyness in place of the underlying and the strike separately) and C/K as target; and $v4$, the same five inputs with $\log(C/K)$ as target. All four are compared on the validation set only, for both underlyings, with the test set left untouched.

Table 5 separates the two transformations that distinguish the four specifications: normalizing the target and the main input by K ($v1, v2 \rightarrow v3, v4$), and taking the

Table 5: Validation-set MAE by specification and moneyness bucket

Moneyness bucket	n	BS	v1 (C)	v2 ($\log(1+C)$)	v3 (C/K)	v4 ($\log(C/K)$)
<i>Panel A: AAPL — XGBoost</i>						
Deep OTM	20,297	0.302	0.243	0.251	5.942	0.208
OTM	21,772	0.835	0.910	0.736	4.893	0.584
ATM	14,114	1.195	1.977	1.797	4.260	1.051
ITM	16,250	0.944	3.600	3.406	2.720	0.929
Deep ITM	32,936	0.496	9.311	9.462	1.400	3.469
Very deep ITM	16,442	0.226	18.377	18.376	4.873	10.462
Extreme ITM	4,144	0.217	27.713	27.698	151.427	100.714
All	125,955	0.615	6.628	6.593	8.616	5.959
<i>Panel B: AAPL — Neural network</i>						
Deep OTM	20,297	0.302	0.690	0.369	11.231	0.367
OTM	21,772	0.835	1.720	1.799	4.222	1.789
ATM	14,114	1.195	2.630	5.327	5.812	5.791
ITM	16,250	0.944	5.059	8.048	2.066	12.770
Deep ITM	32,936	0.496	7.194	6.616	3.026	15.993
Very deep ITM	16,442	0.226	15.724	15.051	1.862	11.138
Extreme ITM	4,144	0.217	20.947	38.338	7.998	93.862
All	125,955	0.615	5.979	6.962	4.755	11.389
<i>Panel C: SPX — XGBoost</i>						
Deep OTM	21,332	2.604	3.078	0.386	0.548	0.571
OTM	157,667	13.499	7.122	6.717	5.835	7.517
ATM	529,060	20.037	24.777	27.603	26.166	15.784
ITM	272,607	24.450	78.571	109.113	122.374	20.274
Deep ITM	143,006	15.348	218.878	316.922	321.679	42.585
Very deep ITM	17,812	5.613	680.560	905.728	804.800	222.304
Extreme ITM	8,241	5.340	1,113.767	1,378.957	1,184.988	1,405.364
All	1,149,725	18.951	76.816	102.837	102.840	31.926
<i>Panel D: SPX — Neural network</i>						
Deep OTM	21,332	2.604	58.706	1.204	76.278	0.642
OTM	157,667	13.499	30.212	9.455	48.644	9.452
ATM	529,060	20.037	37.039	50.768	39.784	47.465
ITM	272,607	24.450	48.811	178.001	22.594	130.610
Deep ITM	143,006	15.348	110.904	210.065	29.837	204.591
Very deep ITM	17,812	5.613	264.283	868.117	33.757	619.913
Extreme ITM	8,241	5.340	755.746	2,031.866	21.457	1,913.448
All	1,149,725	18.951	57.156	121.028	36.138	102.885

Errors in US dollars for AAPL and index points for SPX. All models are evaluated once on the validation set, with hyperparameters as reported in Section 4.5.

logarithm of the target ($v1, v3 \rightarrow v2, v4$). Restricting the comparison to the three buckets away from the in-the-money region, where every specification is already affected by the structural limitations documented in Section 6.1, isolates the two effects more cleanly than the aggregate does.

The two transformations combined (v4) deliver the clearest improvement, though not always relative to the same alternative. Isolating the logarithm alone (v3 to v4, holding the normalised input fixed) produces the largest single gains for the gradient-boosted model: on AAPL deep out-of-the-money the error falls from \$5.94 to \$0.21. Relative to the dollar-scale target under the same log transform (v2), the improvement is far more modest (\$0.25 to \$0.21 on the same bucket), since v2 is already a reasonably strong specification there. Normalisation matters more elsewhere: on SPX at-the-money, v4 reaches 15.78 index points against 27.60 for its dollar-scale counterpart (v2). For the neural network the gain is concentrated but still substantial: v4 is the strongest or tied-strongest specification on both deep out-of-the-money panels, most notably cutting the SPX error from 58.71 index points under the raw target without the log (v1) to 0.64. Neither transformation achieves this improvement on its own.

K -normalization on its own often lowers the error moving into the money: comparing $v1$ to $v3$ and $v2$ to $v4$ in the in-the-money bucket, normalizing improves accuracy in six of the eight model-underlying combinations, the exceptions being the gradient-boosted model on SPX ($v1$ to $v3$) and the neural network on AAPL ($v2$ to $v4$).

The logarithm's effect depends systematically on where in the cross-section it is applied, for both models. It consistently helps deep out-of-the-money and out-of-the-money contracts, where the raw price is small and the log compresses estimation noise, and tends to hurt from the at-the-money bucket onward, where the price is already well approximated by intrinsic value on the linear scale. The pattern is sharpest for the neural network on SPX, where it holds across both comparisons ($v1$ to $v2$, and $v3$ to $v4$): the logarithm cuts the error from \$58.71 to \$1.20 and from \$76.28 to \$0.64 on deep out-of-the-money contracts, but worsens it from the at-the-money bucket onward in both cases, trading in-the-money accuracy for out-of-the-money accuracy rather than improving the fit uniformly.

This is precisely where the improvement matters most for hedging. Combining K -

normalization with the logarithm ($v4$) improves accuracy over the raw specification on deep out-of-the-money contracts in every model-underlying combination, letting the model exploit degree-one homogeneity on the segment of the cross-section where accurate pricing is most consequential for practical use.

$v4$ is retained as the specification used throughout the thesis. For the gradient-boosted model this is a direct consequence of Table 6.2: it is the strongest specification on the out-of-the-money contracts that matter most for hedging, and the model used for the main results in Section 5.1. For the neural network, whose role in this thesis is diagnostic, $v4$ is kept for comparability with the gradient-boosted model rather than because it is the network’s best specification on this evidence, however it is the most accurate on OTM buckets; $v3$ would have been the more accurate choice from the at-the-money bucket onward. This does not affect the diagnostic use made of the network in Section 6.1: the flattening behaviour beyond the training set documented in Figure 4 might follow from the saturation of the hidden units, though whether it is equally pronounced under $v3$ has not been verified directly.

6.3 Robustness to the choice of test period

The comparison in Section 5.1 rests on a single chronological test period. If Black–Scholes’s advantage deep in the money follows from the structure of the estimators, as Section 6.1 argues, it should hold regardless of which period is used to estimate and test the models.

Three additional, non-overlapping rolling splits are constructed from the same AAPL and SPX samples, each with roughly sixteen months of training data, less than half of the ~ 3.5 years used in the main analysis, followed by a four-month test period. Together they span the sample from August 2020 to August 2025 (Table 6). Both machine-learning models are refitted with the hyperparameters already selected in Section 4.5, with no new tuning. Those hyperparameters were selected on data that partly overlap the windows of the additional splits, so the exercise tests the stability of the comparison across periods rather than providing a fully independent out-of-sample evaluation. The neural network’s convergence was confirmed on every split: all runs terminate before the 2,000-epoch limit,

with no evidence of instability from the shorter training windows.

Table 6: Additional temporal splits

Split	Training period	Test period
C	2020-08-31 to 2022-01-01	2022-01-02 to 2022-04-30
B	2022-05-01 to 2023-09-01	2023-09-02 to 2023-12-31
A	2024-01-01 to 2025-05-01	2025-05-02 to 2025-08-29

The deepest in-the-money tail is fully robust. Tables 8 and 9 report the mean absolute error by moneyness bucket for the three additional splits, for the gradient-boosted model and the neural network respectively. Black–Scholes is more accurate than both machine-learning models on very deep and extreme in-the-money contracts in every split for which test observations are available, on both underlyings, with the gap widening toward the deepest bucket in every case. This is the strongest form of confirmation available from this exercise: a result that survives three additional training and test periods, with training samples far smaller than in the main analysis, is unlikely to be an artefact of the particular window used in Section 5.1.

Below the tail, the result holds in direction for the gradient-boosted model on SPX, and less consistently elsewhere. Table 7 counts, for the remaining buckets, how many of the three additional splits confirm the direction found in the main analysis.

For SPX, the gradient-boosted model’s advantage from deep out-of-the-money through in-the-money contracts is confirmed in all three splits, as is Black–Scholes’s advantage in the deep in-the-money bucket. The neural network’s advantage on SPX is confirmed in all three splits only on deep out-of-the-money contracts; on out-of-the-money contracts it holds in one split out of three.

For AAPL, both models lose their advantage in Split C. Since this happens to both machine-learning models at once, it points to a feature of that period rather than of either estimator; its cause is not identified here. On AAPL, the point at which Black–Scholes takes over also varies with the period: in the deep in-the-money bucket, the gradient-boosted model is more accurate than Black–Scholes in two of the three splits, so the crossover moves between moneyness 1.2 and 2. On SPX it stays at 1.2 in every split.

Table 7: Splits (out of 3) confirming the direction of the main result, by bucket

Bucket	AAPL-XGB	AAPL-NN	SPX-XGB	SPX-NN
Deep OTM	2/3	2/3	3/3	3/3
OTM	2/3	2/3	3/3	1/3
ATM	2/3	2/3 ^a	3/3	2/3 ^a
ITM	2/3	3/3 ^a	3/3	3/3 ^a
Deep ITM	1/3 ^a	3/3 ^a	3/3 ^a	3/3 ^a

Unmarked cells: the direction tested is the machine-learning model being more accurate than Black–Scholes, as in the main analysis.

^a The direction tested is Black–Scholes being more accurate, as in the main analysis for that model and bucket.

Full results by bucket and split. Tables 8 and 9 report the underlying mean absolute errors for every bucket and every split, for the gradient-boosted model and the neural network respectively.

Table 8: Mean absolute error by moneyness bucket, three additional temporal splits — gradient-boosted model

Moneyness bucket	Split C		Split B		Split A	
	BS	XGB	BS	XGB	BS	XGB
<i>Panel A: AAPL</i>						
Deep OTM	0.217	1.351	0.120	0.103	2.143	0.171
OTM	0.483	2.166	0.418	0.384	2.930	0.697
ATM	0.855	1.885	0.950	0.848	3.422	1.275
ITM	1.007	1.112	1.047	0.647	2.806	0.933
Deep ITM	1.016	0.795	0.546	1.335	1.235	1.143
Very deep ITM	0.398	2.234	0.177	4.163	0.282	3.249
Extreme ITM	0.285	5.896	—	—	0.262	18.347
<i>Panel B: SPX</i>						
Deep OTM	7.110	1.020	0.777	0.312	45.055	0.635
OTM	15.046	9.248	5.969	4.539	76.943	9.118
ATM	19.556	18.822	14.321	10.394	57.170	21.263
ITM	29.136	16.465	18.659	11.955	46.292	13.948
Deep ITM	28.616	32.055	12.474	54.986	17.195	36.507
Very deep ITM	7.474	164.615	2.799	221.710	4.167	363.366
Extreme ITM	5.796	1,176.532	2.392	1,092.294	3.554	1,505.523

Errors in US dollars for AAPL and index points for SPX. “—” denotes a bucket with no test observations in that split.

Interpretation. The two halves of the reversal documented in Section 5.1 behave differently across periods. Black–Scholes’s advantage on very deep and extreme in-the-money contracts does not depend on the period: it is the most robust finding in this thesis.

Table 9: Mean absolute error by moneyness bucket, three additional temporal splits — neural network

Moneyness bucket	Split C		Split B		Split A	
	BS	NN	BS	NN	BS	NN
<i>Panel A: AAPL</i>						
Deep OTM	0.217	0.816	0.120	0.051	2.143	0.150
OTM	0.483	0.720	0.418	0.272	2.930	0.534
ATM	0.855	1.421	0.950	1.464	3.422	1.796
ITM	1.007	3.350	1.047	2.875	2.806	7.041
Deep ITM	1.016	4.842	0.546	2.652	1.235	22.566
Very deep ITM	0.398	10.125	0.177	4.652	0.282	23.689
Extreme ITM	0.285	36.268	—	—	0.262	54.977
<i>Panel B: SPX</i>						
Deep OTM	7.110	1.559	0.777	0.628	45.055	1.386
OTM	15.046	17.717	5.969	6.403	76.943	15.188
ATM	19.556	91.668	14.321	31.692	57.170	50.358
ITM	29.136	270.247	18.659	62.031	46.292	125.864
Deep ITM	28.616	278.521	12.474	152.356	17.195	440.814
Very deep ITM	7.474	176.634	2.799	623.969	4.167	2,492.156
Extreme ITM	5.796	1,066.216	2.392	1,265.741	3.554	2,486.688

Errors in US dollars for AAPL and index points for SPX. “—” denotes a bucket with no test observations in that split.

This is consistent with the diagnosis in Section 6.1, which traces it to how the estimators represent that region (the tree’s constant-valued terminal leaves and, for the network, the log-transformed target) rather than to the data of a particular window.

The advantage below the tail is stable in direction for the gradient-boosted model on SPX, but its size varies considerably, and the variation comes almost entirely from Black–Scholes. On SPX deep out-of-the-money contracts, the gradient-boosted model’s error stays between 0.31 and 1.02 index points across the three splits, while Black–Scholes’s error ranges from 0.78 to 45.06. The size of the machine-learning advantage therefore depends on how far the historical volatility supplied to Black–Scholes is from the level the market prices in during the test period, a feature of market conditions rather than of either model. The robustness check is consistent with this distinction between the two parts of the result, though three additional splits are not enough to establish it.

6.4 Limitations and future work

For all models, volatility and the risk-free rate are not matched to the maturity of each contract. Every contract receives the same 60-day historical volatility and the rate at the 90-day tenor, whatever its maturity (Section 4.3). Since all models receive the same inputs, the comparison is not affected, but matching both inputs to each contract’s maturity could improve the accuracy of all of them. This is common practice in the literature comparing machine-learning and Black–Scholes pricing, but it is a simplification: in the actual test data, moneyness, maturity and volatility vary jointly, and a model’s behaviour beyond the training range may depend on combinations of all three rather than on moneyness alone.

Neither machine-learning model respects the no-arbitrage bounds that a call price must satisfy beyond the training range: Figure 4 shows the gradient-boosted model’s prediction exceeding the underlying price, and the neural network’s prediction falling below intrinsic value, at moneyness levels rarely observed in the sample (fewer than 1% of training contracts lie beyond moneyness 5).

The neural network’s architecture was kept deliberately simple: a single hidden layer, with the number of units selected on a single train/validation split rather than through cross-validation, a choice made for computational reasons and declared as a limitation in Section 4.5. The network was trained on CPU rather than GPU, which constrained both the size of the hyperparameter search and the number of training epochs feasible within the time available; some of the network’s weaker performance relative to the gradient-boosted model, particularly in the in-the-money region documented in Section 5.1, may reflect this rather than a fundamental limit of neural networks as a function class. The universal approximation theorem guarantees that a network with a single hidden layer can approximate any continuous function on a compact domain arbitrarily well, given enough hidden units and sufficient training; the network used here was not tuned aggressively enough, nor trained long enough, to be considered a test of that ceiling.

A natural extension is therefore a more thorough tuning of the neural network, including a proper cross-validated search over the number of hidden units, and experimentation with deeper architectures or a different activation function. In particular, the flattening

behaviour documented in Figure 4 may be related to the saturation of the logistic activation function used in the hidden layer: each hidden unit is bounded between 0 and 1, so the network’s prediction is bounded to converge to a constant far from the training data, regardless of the linear behaviour the data would imply beyond the observed range. An activation function that does not saturate on one side, such as ReLU, would not impose this constraint and could in principle extrapolate linearly beyond the training boundary; whether this would also address the deep in-the-money failure documented in Section 5.1 is left for future work.

A second direction follows directly from the deep in-the-money failure diagnosed in Section 5.1: rather than have the model learn the full price C , it could learn only the time value $C - \max(S - K, 0)$, adding the intrinsic value back analytically afterwards. Time value tends to zero as an option moves deep in the money, so even a model that flattens beyond the training range would flatten toward a quantity that is itself close to zero there, and the reconstructed price would inherit the correct linear behaviour from the analytically added intrinsic value rather than from the estimated component. This would not address the out-of-the-money region, where time value is the entire price, but could remove the deep in-the-money failure that both models currently share.

7 Conclusion

This thesis set out to compare Black–Scholes against two machine-learning models, XG-Boost and a neural network, for pricing options on AAPL and SPX. The comparison used a common information set, a chronological split, and an evaluation by moneyness to identify where each model performs well and where it falls short.

Section 5.1 showed a systematic reversal across both underlyings: the gradient-boosted model outperformed Black–Scholes on out-of-the-money, at-the-money and moderately in-the-money contracts, while Black–Scholes was more accurate for deeply in-the-money contracts, with its advantage widening further into the money. The neural network, in the specification used for the main comparison, improved on Black–Scholes only in the out-of-the-money buckets. Aggregate errors concealed these differences, since they also reflected the composition of each sample.

Section 6.1 examined possible explanations for this pattern. The high in-sample errors in the extreme in-the-money region indicate that the failure cannot be attributed solely to poor generalisation. The synthetic exercise in Figure 4 also highlighted the piecewise-constant predictions of the tree ensemble and the flattening behaviour of the neural network with logistic activation. These specifications do not enforce the pricing relationships and limiting behaviour that Black–Scholes incorporates by construction. Their performance nevertheless also depends on the target transformation, the distribution of the training data and the volatility estimate supplied to the benchmark.

These findings reinforce the methodological concerns discussed by Ruf and Wang (2020): the measured advantage of machine learning depends on the information supplied to each model, the evaluation design and the region of the contract space considered. The practical implication is that pricing accuracy should be assessed alongside economic consistency, particularly in regions sparsely represented in training. Within the sample studied here, flexible estimators offer improvements in specific regions, while the deep in-the-money results suggest that incorporating known pricing structure into their predictions is a useful direction for further work.

A Raw data fields

Table 10: Fields used from the dividend distribution file (AAPL)

Field	Role
<code>ex_date</code>	Date from which the underlying trades without entitlement to the distribution; used as the effective date for the as-of merge onto the option sample.
<code>amount</code>	Cash amount of the distribution, in dollars per share, when <code>distr_type</code> = 1; retained only for these declared regular dividends, not for the yield projections Option-Metrics reports under <code>distr_type</code> = %.
<code>adj_factor</code>	Adjustment applied to compare pre- and post-distribution prices; used to identify the stock-split record, rather than a cash distribution, in the file.
<code>distr_type</code>	Distribution type. Filtered to 1 (regular dividend); this excludes splits, stock dividends, capital gain distributions, special dividends, spin-offs, and yield projections.
<code>frequency</code>	Declared payment frequency (quarterly for AAPL throughout the sample); used to annualise the most recent declared dividend into a yield.
<code>cancel_flag</code>	Distinguishes distributions paid as scheduled from those cancelled or omitted. 0 – The distribution was or will be made as scheduled; 1 – The distribution was cancelled, or a regular payment was omitted.
<code>liquid_flag</code>	Distinguishes ordinary distributions from partial or total liquidating distributions. 0 – The distribution is a non-liquidating distribution File Formats IvyDB US Reference; 1 - the distribution is either a partial or total liquidating distribution

Table 11: Fields in the raw historical volatility files (`*_historical_volatility.csv`) and their treatment

Field	Kept	Description	AAPL / SPX detail
<code>secid</code>	No	Security identifier	Constant within each file; used only to confirm the correct security was returned by the extract
<code>date</code>	Yes	Date the volatility is computed as of	Merge key
<code>days</code>	Yes	Length of the trailing window, in calendar days	Ranges from 10 to 1825; only <code>days = 60</code> is retained
<code>volatility</code>	Yes	Realised volatility over the trailing window	Standard deviation of the logarithm of close-to-close daily total returns, annualised; becomes σ
<code>index_flag</code>	No	Index underlying indicator	0 for AAPL, 1 for SPX; consistency check only

Notes. The file reports thirteen values of `volatility` per date, one per window length. The construction and limitations of σ are discussed in Section 2.2.

Table 12: Fields of the risk-free rate file (`riskfree_rate.csv`)

Field	Kept	Description	AAPL / SPX detail
<code>date</code>	Yes	Date of the zero-coupon curve.	Merged with both option samples on the quote date.
<code>days</code>	Yes	Tenor of the curve point, in calendar days.	Tenors are not on a fixed grid: each date has between 11 and 46 points (7 to 3,657 days). The point closest to 90 days is selected on each date.
<code>rate</code>	Yes	Zero-coupon interest rate for the given tenor, in percent. ^a	Same series for both underlyings; divided by 100 before use.

^a Annualized, continuously compounded rate as provided by OptionMetrics IvyDB.

Table 13: Fields in the raw option-price files (*_options_prices.csv) and their treatment

Field	Kept	Description	AAPL / SPX detail
date	Yes	Quotation date	Merge key; 1,256 trading days in both files
exdate	Yes	Expiry date	Used to compute T
cp_flag	No	Call (C) or put (P)	Filtered to C; 997,347 of 1,994,694 rows retained for AAPL, 12,235,448 of 24,470,893 for SPX; column then dropped as constant
strike_price	Yes	Strike, $\times 1000$	Divided by 1000 before use
best_bid	Yes	Highest quoted bid	Rows with a non-positive bid dropped: 83,947 for AAPL, 587,254 for SPX
best_offer	Yes	Lowest quoted ask	Enters the target together with best_bid
volume	No	Contracts traded that day	Used with open_interest to drop untraded contracts (65,372 rows for AAPL, 3,841,638 for SPX); not supplied as a model input
open_interest	No	Contracts outstanding	As above
optionid	No	Contract identifier	17,223 distinct contracts for AAPL, 211,827 for SPX; used to motivate the chronological split in Section 4.4
cfadj	No	Contract adjustment factor	Constant 4 for AAPL until 16 Sep 2022 (four-for-one split of 31 Aug 2020), 1 thereafter; strikes already adjusted at source, so no rescaling applied; constant 1 for SPX
ss_flag	No	Settlement type	Constant 0 in both files
index_flag	No	Index underlying indicator	0 for AAPL, 1 for SPX
issuer	No	Issuer name	Constant within each file
exercise_style	No	American (A) or European (E)	A for AAPL, E for SPX; motivates the treatment of assumption (A5) in Section 2.3

Notes. Row counts refer to the option-price files after each successive filter, in the order applied (Section 4.2). Fields from the dividend file are listed separately in Table 10; the underlying's closing price, the risk-free curve, and historical volatility are described in Section 4.3 and drawn from separate files.

Table 14: Hyperparameter grids

Hyperparameter	Values searched		Selected	
	AAPL	SPX	AAPL	SPX
<i>XGBoost (AAPL: 72 combinations; SPX: 18 combinations)</i>				
<code>max_depth</code>	5, 7, 9, 10	3, 5, 7	10	7
<code>learning_rate</code>	0.01, 0.05, 0.10	0.01, 0.05, 0.10	0.10	0.10
<code>n_estimators</code>	100, 300, 500	early stopping ^a	500	87
<code>subsample</code>	0.8, 1.0	0.8, 1.0	0.8	1.0
<i>Neural network</i>				
Hidden units	1–20	1–15	10	10

XGBoost: selection across five rolling folds (24-month training, 6-month validation, 6-month step), using the root mean squared error on the price scale: among the configurations whose mean error is within 1% of the lowest, the one with the smallest standard deviation across folds is retained. Target $\log(C/K)$; all other settings at library defaults. Neural network: selection by root mean squared error on the price scale on a single chronological validation set, retaining the number of units after which the error stops improving materially. Fixed settings: one hidden layer, logistic activation, Adam optimiser, mini-batch size 2,000, L2 penalty 10^{-4} , at most 2,000 epochs, early stopping after 15 epochs without improvement on a random 10% of the training data. Target $\log(C/K)$, inputs standardised.

^a At most 1,000 trees, stopped after 30 rounds without improvement on the validation fold. For the final model, the number of trees is set by early stopping on the last 10% of the training and validation period.

Table 15: Number of observations by moneyness bucket, estimation and test periods

Moneyness bucket	AAPL			SPX		
	<i>n</i> train	<i>n</i> val	<i>n</i> test	<i>n</i> train	<i>n</i> val	<i>n</i> test
Deep OTM	105,285	20,297	26,376	155,781	21,332	30,051
OTM	95,876	21,772	22,324	1,035,010	157,667	227,962
ATM	73,909	14,114	13,178	2,184,122	529,060	490,931
ITM	87,994	16,250	15,059	1,250,438	272,607	251,452
Deep ITM	144,521	32,936	31,571	587,889	143,006	123,621
Very deep ITM	74,966	16,442	11,824	130,570	17,812	17,799
Extreme ITM	4,553	4,144	4,885	44,271	8,241	8,528
All	587,104	125,955	125,217	5,388,081	1,149,725	1,150,344

Train and validation refer to the estimation periods used for the rolling-window cross-validation; the final model in each case is refitted on the two combined.

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